

Session 1.4: Tabular Data Wrangling Part 2: Advanced Filtering, Multi-Segment GroupBy, and Financial Auditing

PEDAGOGICAL AGENDA

1:2 Delivery Rhythm

Session Learning Architecture

Structured 3-hour applied computing session designed for business students with zero prior coding experience.

- **Phase 1 (60 Min Lecture):** Computational foundations, business vocabulary, and syntactic mechanics.
- **Phase 2 (120 Min Lab):** Guided hands-on implementation, error diagnosis, and AI co-pilot resolution.
- **Zero Fluff Mandate:** Focus directly on executable Python code, tabular data wrangling, and commercial intelligence.

STRATEGIC OUTCOME

Operational Autonomy

Business Analytical Competency

Transforming passive spreadsheet users into autonomous programmatic decision makers.

- **Speed & Scale:** Processing datasets exceeding spreadsheet limits with instantaneous vectorized execution.
- **Repeatable Pipelines:** Replacing manual copy-paste workflows with fully automated Python scripts.
- **AI Debugging Literacy:** Leveraging modern IDEs to diagnose and resolve errors independently.

Core Computational Keywords & Business Meaning

CONCEPT 01

Core Principle

Syntactic Rigor

Writing clean, self-documenting code that scales across corporate systems.

CONCEPT 02

Core Principle

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CONCEPT 03

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CONCEPT 04

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Computational Foundations & Business Operations

PILLAR 01

Architecture

Boolean Indexing at Enterprise Scale

In business reporting, filtering data across multiple conditions requires bitwise operators (`&`, `|`) because standard Python `and`/`or` operate on single scalar booleans, whereas Pandas evaluates element-wise boolean arrays.

PILLAR 02

Architecture

The Split-Apply-Combine Paradigm

Grouping data by corporate dimensions (e.g., `Division_Name`, `Fiscal_Quarter`) to calculate sub-totals and unit economics without manually building pivot tables.

PILLAR 03

Architecture

Relational Joins vs. Fragile Spreadsheet Lookups

Spreadsheet lookup formulas fail when columns are rearranged or when duplicate keys exist. A Pandas merge (`how="inner"`, `how="left"`) explicitly specifies match criteria and logs unmapped rows automatically.

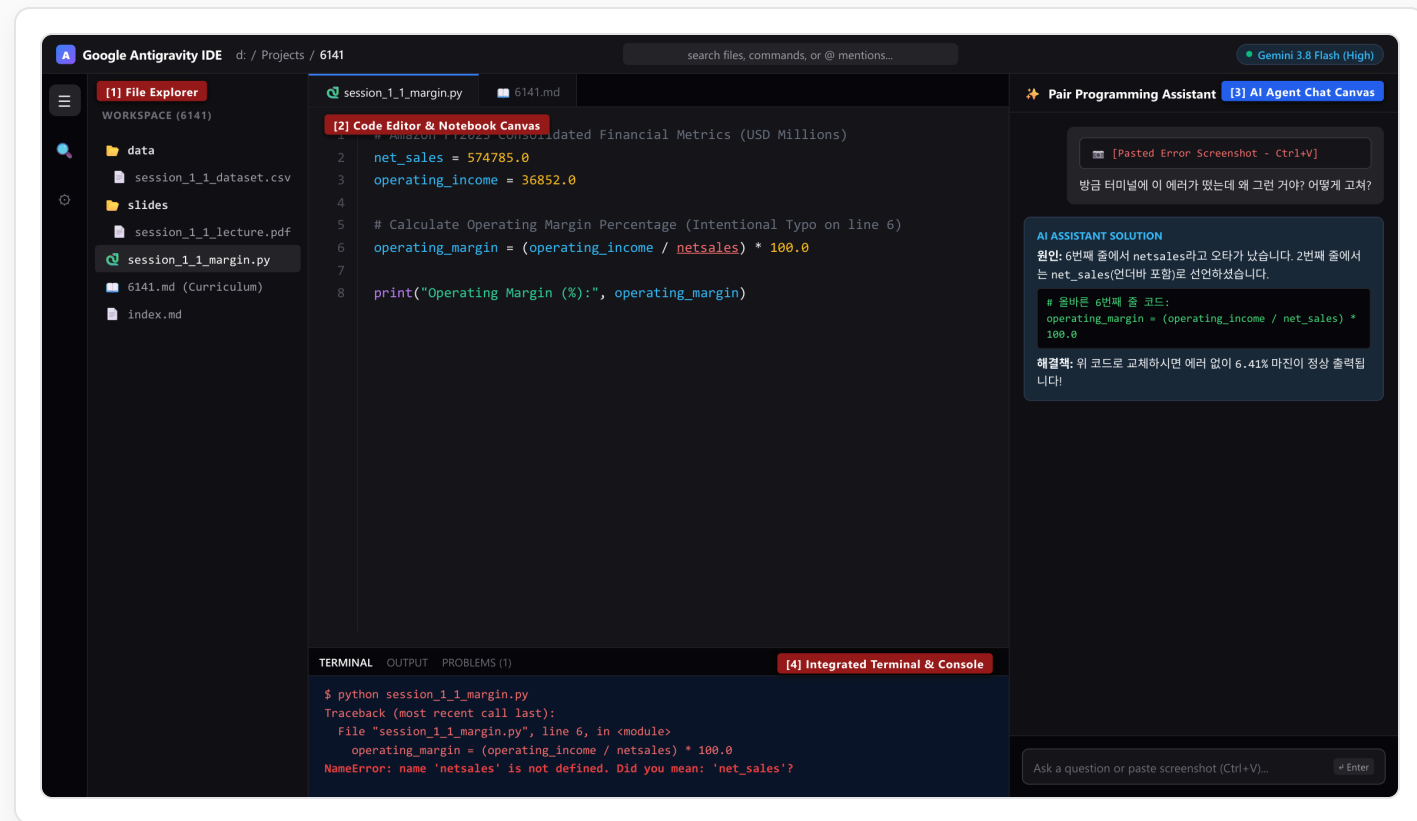
PILLAR 04

Architecture

Automated Audit Exception Flags

Constructing conditional categorical flags (e.g., `Variance_Flag` when `Actual_Expense > Budget_Expense * 1.05`) to route financial discrepancies directly to corporate controllers.

Google Antigravity IDE Architecture & AI Debugging Protocol



[1] File Explorer

Project Hierarchy

Manage datasets in data/, scripts in root, and exported artifacts.

[2] Code Editor

Syntax Workspace

Modern syntax highlighting, lint diagnostics, and line-by-line script authoring.

[3] AI Agent Chat

AI Pair Programmer

Paste error screenshots directly to inspect root causes and get verified fixes.

[4] Terminal

Execution Engine

Run Python scripts (python script.py) and view console logs and tracebacks.

Zero-Friction AI Error Resolution Protocol

- Capture:** Press Win + Shift + S to clip terminal error traceback.
- Paste:** Press Ctrl + V into AI Chat [3] with "Explain root cause & fix".
- Apply:** Review the explanation and apply verified fix into Editor [2].

Script Demonstration 1: Script Demonstration 1

```
session_1_4_demo_1.py Python 3.10

1  "kw">class="kw">import pandas "kw">class="kw">as pd
2
3      df = pd."fn">read_csv("data/session_1_4_dataset.csv")
4      mask = (df["Division_Name"] == "Consumer") & (df["Actual_Expense_USD"] > df
5      overrun_txns = df[mask]
6      "fn">print(f"Consumer Overrun Transactions Count: {">len(overrun_txns)}")

$ python session_1_4_demo_1.py [Exit 0]
```

1 Syntactic Structure

Importing production libraries and defining clean analytical data structures.

2 Vectorized Transformation

Applying deterministic operations across the full dataset without fragile cell references.

3 Console Verification

Emitting formatted business KPIs to the terminal for auditability and compliance.

Script Demonstration 2: Script Demonstration 2

session_1_4_demo_2.pyPython 3.10

1

div_summary = df."fn">groupby("Division_Name")."fn">agg(
2

Total_Budget_USD=("Budget_Expense_USD", "sum"),
3

Total_Actual_USD=("Actual_Expense_USD", "sum"),
4

Transaction_Count=("Transaction_ID", "count")
5

)
6

div_summary["Variance_USD"] = div_summary["Total_Actual_USD"] - div_summary
7

div_summary["Variance_Pct"] = (div_summary["Variance_USD"] / div_summary["To
8

"fn">print(div_summary[["Total_Actual_USD", "Variance_USD", "Variance_Pct"]]

\$ python session_1_4_demo_2.py [Exit 0]

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Applied Code Walkthrough | Style 2 Harvard Academic Standard

Executable Python Script

Slide 6 / 8

Script Demonstration 3: Script Demonstration 3

session_1_4_demo_3.pyPython 3.10

```
1 dept_info = pd.DataFrame({
2     "Cost_Center_Code": ["CC-101", "CC-202", "CC-303", "CC-404"],
3     "VP_Lead": ["Sarah Chen", "Marcus Vance", "Elena Rostova", "David Kim"]
4 })
5 merged_ledger = pd.merge(df, dept_info, on="Cost_Center_Code", how="left")
6 "fn">print(merged_ledger[["Transaction_ID", "Division_Name", "VP_Lead", "Ac"]])
```

\$ python session_1_4_demo_3.py [Exit 0]

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Interactive Two-Tier Lab Protocol & AI Diagnosis

EXERCISE 1

Guided Lab

Foundational Implementation

Step-by-step guided practice establishing baseline syntax mastery in Antigravity IDE.

- Step 1:** Ingest dataset from `data/session_1_4_dataset.csv`.
- Step 2:** Implement core analytical functions and compute primary business metrics.
- Step 3:** Format console outputs and verify calculations against baseline ledger totals.
- Step 4:** Run in Terminal [4] and confirm clean execution with exit code 0.

EXERCISE 2+

Coding Challenge

Advanced Scripting & AI Debugging

Applied programming challenge expanding pipeline logic and resolving intentional exceptions.

- Task 1 (Pipeline Expansion):** Add secondary business unit logic and multi-tier calculations.
- Task 2 (Deliberate Error & AI Capture):** Introduce intentional syntax or type error, capture traceback with `Win + Shift + S`, and diagnose via AI Chat [3].
- Task 3 (Verified Production Run):** Apply verified fix, export audit output, and confirm zero errors with return code 0.